

**MCAST**

Malta College of Arts, Science & Technology

INSTITUTE OF INFORMATION
AND COMMUNICATION TECHNOLOGY

Low Level Programming

1BSc Software Development

Assignment 1 (Class)

Low Level Programming

Instructions to Students

Read the following instructions carefully before you start the assignment.

- This assignment is a class assignment and must be completed in **2 hrs**.
 - This assignment has a total weight of **51%**.
 - ALL questions must be attempted.
 - Copying is strictly prohibited and will be subjected to disciplinary procedures as per MCAST Policies, Procedures and Regulations.
 - **Use of any electronic equipment, including calculators, during this exam is strictly prohibited.**
 - Write your full name on the cover sheet of the assignment.
-

KU1	Outline specific commands used to implement pre-processor directives
-----	--

1. (a) Write down a preprocessor directive in C to define a constant PI equal to 3.142 **[1]**

- (b) The following macro multiplies 2 numbers.

```
#define MULTIPLY(x, y) (x) * (y)
```

Work out the value of z after the following code is executed. **[2]**

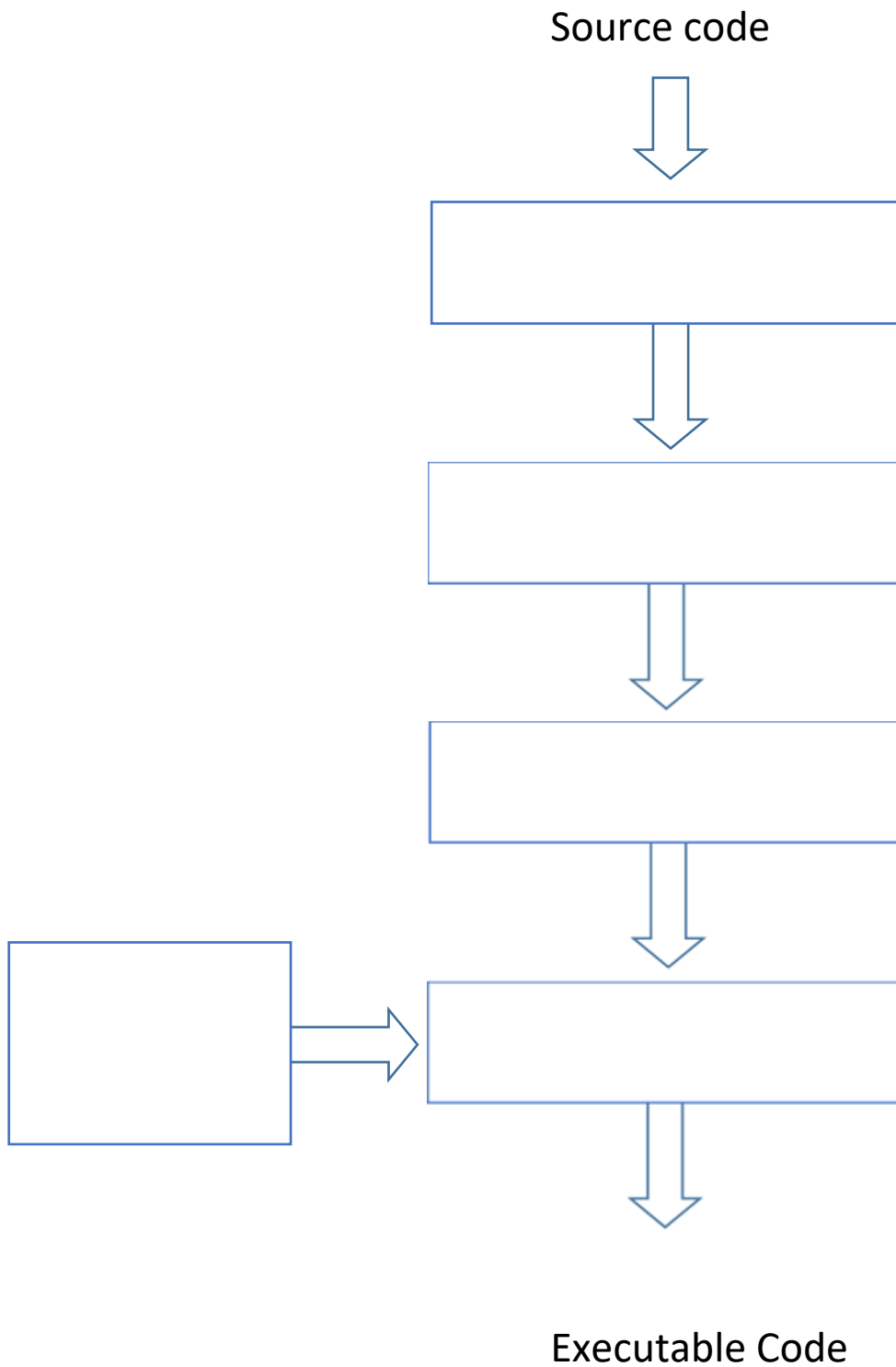
```
int z= MULTIPLY(5-2, 2);
```

Why are the brackets around x and y required in the definition of the macro? **[2]**

KU2	Show an understanding of the 'C' language Compiling and Linking process
-----	---

2. (a) Fill in the diagram below with the terms provided to show the order of compilation of a program in C. [5]

Terms: Linking, Assembly, Compilation, Pre-processing, Library Files



KU4	Identify suitable data types for a given scenario
-----	---

3. (a) In the space provided below, write a code snippet in C to

- declare an integer, **num** and store the value 10
- set a pointer, **p** to point to **num**
- display the contents of **num** by using the pointer, **p**

[1, 1, 1]

(b) Suppose that the integer **num** has a value of 10 and is currently stored at memory location 1000. What is the value stored in the pointer, **p** after it is set to point to **num**? [1]

(c) What is the new value stored in the pointer, **p** after the following code is executed? [1]

`p++;`

KU5	Outline commands to handle strings in C
-----	---

4. (a) Write code in C to show how a string can be declared in C using two different methods. [1]

- (b) For a string to be a valid C string it must be terminated with a special character. [1]
Which is this character?

- (c) In C, two string variables cannot be compared using the == operator. [1, 1]
Why? How should strings be compared in C?

- (d) **scanf()** is considered to be unsafe in C because of the buffer overflow problem. [1]
Suggest one way how this problem can be solved when inputting a string.

KU6	Outline commands to implement conditional statements
-----	--

5. (a) What is the output of this program?

[2]

```
#include <stdio.h>
int main(void) {
    int i = 5;
    if(!i)
        i = 2;
    else
        i = 3;
    printf("%d\n",i);
    return 0;
}
```

(b) Re-write the following ternary expression using an if() statement in the space provided.

[3]

```
big = a > b ? (a > c ? a : c) : (b > c ? b : c);
```

AA2	Produce relevant 'C' code to implement bitwise data manipulation
-----	--

6. (a) What bitwise operator and mask are required to set the 3rd bit of an 8-bit pattern?
For example, 0100 1110 becomes 0110 1110. [1]

Operator: _____

Mask (in binary): _____

- (b) Which bitwise operator is used to check whether two bit patterns are the same? [1]

- (c) Write down the bitwise operation to **invert** the 2nd and 4th bit of an 8-bit pattern stored in a character variable, **ch**. (You may leave the mask in binary) [2]

For example, 0100 1110 becomes 0001 1110

- (d) How can you obtain the original character in (c)? [2]

- (e) What is the technique in (c) and (d) used for? [1]

AA3	Produce relevant 'C' code to implement arithmetic operations
-----	--

7. (a) What is the value of the Z variable at the end of the following snippet? **[2]**

```
int X = 1;
int Y = X + 2;
Y++;
int Z = Y;
--X;
Y--;
Z = X / Y * X * Y;
```

- (b) Assuming that the size of an integer is 4 bytes, what is the output of this program? **[2]**

```
#include <stdio.h>
int main()
{
    int i = 5, j = 10, k = 15;
    printf("%d ", sizeof(k /= i + j));
    printf("%d", k);
    return 0;
}
```

- (c) In the space provided below write a code snippet in C to display the numbers from 1 to 1000, one below the other. The message 'Press any key to continue..' should be displayed after every 25 numbers. **[3]**

AA4	Produce relevant 'C' code to implement data management by using suitable data structures
KU7	Outline commands to implement iterative statements

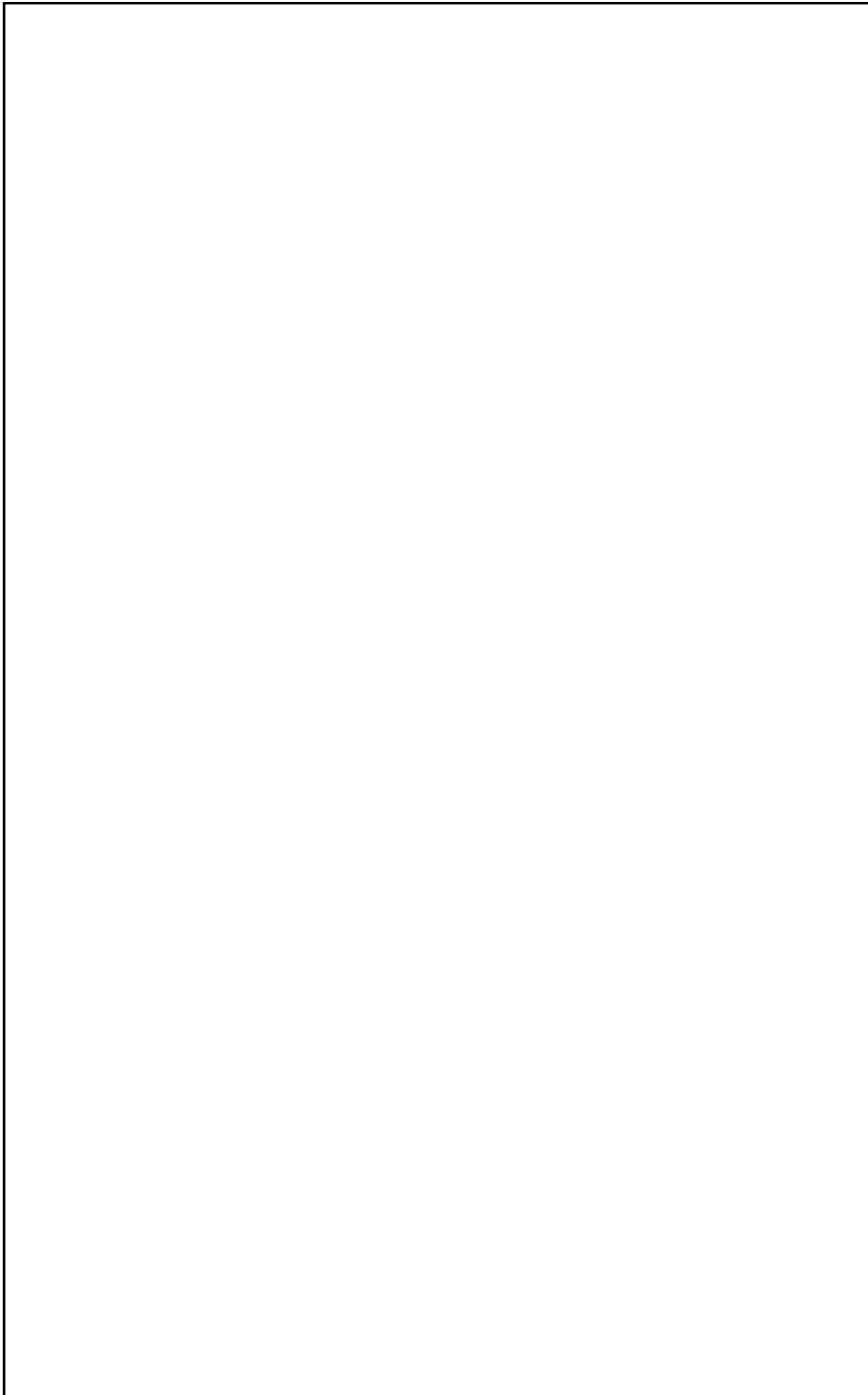
8. A program is required to store the studentId, firstname and mark of students in a class.

- (a) In the space below, write a code snippet in C to **declare a data structure** to store **one student record**. [2]

- (b) In the space below, write a code snippet in C to **declare a data structure** to store **up to 100 student records**. [2]

- (c) In the space below, write a code snippet in C to ask the user to enter the details of ONE student and store it in the data structure declared in (a). [3]

- (d) In the space provided, write a code snippet in C to prompt the user to enter the details of some students one record at a time and store them in the data structure declared in (b). After each record is stored, the user must be prompted to check whether more records need to be entered. The process continues until the user decides to stop or the data structure is full. **[5]**



FINAL MARK			
KU1	Question 1 (a) correct code to define pre-processor directive	1	
	Question 1 (b) correct value of z after calling MULTIPLY()	2	
	Question 1 (b) correct explanation for parenthesis in macro	2	
		5	

KU2	Question 2 One mark for each term placed correctly	5	
------------	--	----------	--

KU4	Question 3 (a) Correct code to define, set and display	3	
	Question 3 (b) Value stored in pointer is correct	1	
	Question 3 (c) New value after pointer is incremented is correct	1	
		5	

KU5	Question 4 (a) Half mark for each correct string declaration	1	
	Question 4 (b) Correct end of string character	1	
	Question 4 c) Correct reason to why strings cannot be compared	1	
	Question 4 c) Method suggested to compare strings is correct	1	
	Question 4 (d) One correct suggestion to fix buffer overflow problem with scanf()	1	
		5	

KU6	Question 5 (a) Correct output	2	
	Question 5 (b) Correct if() statement instead of ternary	3	
		5	

KU7	Question 8(d)		
	Repetition control structure to fill in data structure is correct with suitable boundaries	3	
	Data is properly stored in data structure	1	
	User is prompted to check whether more records are required and input accepted	1	

		5	
--	--	----------	--

AA2	Question 6 (a) Half mark each for correct operator and mask	1	
	Question 6 (b) correct bitwise operator to check for identical patterns	1	
	Question 6 c) correct bitwise operation to invert bits	2	
	Question 6 (d) explanation on how to obtain original character is correct	2	
	Question 6 e) technique identified correctly	1	
		7	

AA3	Question 7 (a) Correct value of Z	2	
	Question 7 (b) Correct output of program	2	
	Question 7 c) Code provided must meet requirements to display numbers 25 at a time	3	
		7	

AA4	Question 8 (a) data structure to store one record properly declared	2	
	Question 8 (b) data structure to store up to 100 records properly declared	2	
	Question 8 c) code to prompt user, accept input and store data in suitable data structure for one record is correct	3	
		7	

	TOTAL:	51	
--	---------------	-----------	--